

Erdős Problem #819

EP-819

L1 - TRACTABLE

PARTIALLY SOLVED

Combinatorics

ORIGINAL PROBLEM STATEMENT

Let $f(N)$ be maximal such that there exists $A \subseteq \{1, \dots, N\}$ with $|A| = \lfloor N^{1/2} \rfloor$ such that $|(A+A) \cap [1, N]| = f(N)$. Estimate $f(N)$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>